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Machine learning land surface temperature downscaling method based on Landsat 9 and Sentinel-2 satellite feature interaction

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ABSTRACT

As a core essential climate variable (ECV) in the Global Climate Observing System, land surface temperature (LST) plays a pivotal role in climate change monitoring, urban thermal environment assessment, agricultural management, and ecosystem surveillance. To obtain high-precision LST data, a novel machine learning-based LST downscaling framework integrating feature interaction optimization and Shapley additive explanations (SHAP) scoring was proposed. SHAP scoring was employed for feature importance analysis to identify optimal predictors, while 10 distinct models were comparatively evaluated to establish a high-resolution downscaling framework adaptable to homogeneous surface characteristics. The results show that the SHAP-based feature selection significantly enhanced prediction accuracy by prioritizing nonlinear determinants. The red-blue band interaction feature demonstrated consistent dominance across all algorithms (XGBoost, LightGBM, GradientBoost), exhibiting both the broadest SHAP value range (−2.0 to 2.0) and the highest relative contribution weight. By explicitly addressing spatial heterogeneity, the spatial random forest (SRF) model achieved superior downscaling performance, particularly in vegetated regions. It generated reliable 10 m-resolution LST estimates ($R^2 = 0.74$, RMSE = 6.28°C), demonstrating robust generalization capabilities in complex terrain conditions. The SHAP-based land surface temperature downscaling method can effectively capture the nonlinear interactions among spectral, topographic, and other features, demonstrating high accuracy and strong physical interpretability in high-resolution temperature retrieval over areas dominated by homogeneous vegetation.

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Land surface temperature; downscaling; machine learning; Landsat 9; Sentinel-2; SHAP

1. Introduction

Land surface temperature (LST), one of the essential climate variables (ECVs) identified by the Global Climate Observing System (GCOS), plays a critical role in the Earth system's energy balance and hydrological cycle (Radeloff et al. 2024). Accurate LST information is of great significance in fields such as climate change monitoring (Chiara, Francesco, and Nicoletta 2024), urban surface thermal environment assessment (Shahfahad et al. 2024), precision agriculture (Ghafarian et al. 2024), and ecosystem monitoring (Zhang, Kasimu, et al. 2024). Thermal infrared (TIR) remote sensing serves as a pivotal methodology for regional-scale LST observation. Contemporary TIR sensors, such as ECO system spaceborne thermal radiometer experiment on space station (ECOSTRESS) (70 m) and advanced spaceborne thermal emission and reflection radiometer (ASTER) (90 m), provide moderate spatial resolution datasets (Hu et al. 2023). However, ECOSTRESS has a revisit cycle of 3–5 d (Hu et al. 2022), whereas ASTER's temporal resolution extends to 16 d (Hulley and Hook 2011), compromising their capacity to capture rapid LST dynamics. The current LST products suffer from limited spatial and temporal resolutions, restricting fine-grained decision-making in urban heat management (Hu et al. 2025; Wang, Dai, Zhang, Chen, et al. 2025; Wang, Dai, Zhang, Chen, Shu, et al. 2025), climate monitoring (Zhang, Tan et al., 2024), and ecological protection (Zhang, Yiğit et al., 2024). These applications require LST products with both high spatial and temporal resolution. However, the effectiveness of current LST downscaling approaches largely depends on model structures and parameterization strategies. At present, both aspects face notable limitations, leading to unsatisfactory performance in capturing fine-scale thermal patterns. In particular, existing methods often struggle with nonlinear feature representation and the integration of multi-source predictors, which restricts their ability to generate reliable high-resolution LST products.

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Generally, LST downscaling can be categorized into three main approaches: empirical/semi-empirical methods, geographically weighted regression (GWR) methods, and machine learning-based methods. Empirical/semi-empirical is computationally efficient and straightforward to implement (Bonafoni 2016; Zhang, Zhao, and Yang 2019). However, due to their dependence on regional-specific empirical relationships, their general applicability is limited, particularly in areas with heterogeneous land cover types. Geographically weighted regression methods address the spatial heterogeneity by incorporating location-specific weighting mechanisms, effectively capturing the localized relationships between LST and environmental variables (Wang, Luo, and Peng 2020; Wang et al. 2021). However, their downscaling accuracy is often constrained to the 100-m scale, with additional limitations in model complexity and potential overfitting risks. Machine learning methods, particularly deep learning architectures, have demonstrated significant advantages in LST downscaling due to their robust capacity for feature extraction and nonlinear modeling (Liang et al. 2023; Wu and Li 2019). These methods automatically discover complex relationships among multi-source datasets and maintain strong generalization ability. Nevertheless, challenges remain, such as effective integration of spatial contextual information, optimization of computational efficiency without compromising model performance, and enhancement of model interpretability, all of which require further investigation.

The performance of machine learning models in LST downscaling is constrained by two critical factors: feature variable selection and the applicability of the models. The diversity and rationality of feature variables play a crucial role in LST downscaling. The most commonly used feature variables can be divided into four main categories (Bonafoni 2016; Guo et al. 2024): spectral indices reflect surface energy absorption and radiation characteristics (e.g. RGB, NIR, and SWIR), remote sensing vegetation indices capture the cooling effects of evapotranspiration (e.g. NDVI, MNDWI, MSAVI, etc.) (Ha, Gowda, and Howell 2011; Liang et al. 2023; Wu et al. 2019; Xu et al. 2021), urbanization indices describe the heat accumulation from impervious surfaces (e.g. NDBI, BI, IBI) (Guo, Hu, and Schlink 2022; Wang, Luo, and Peng 2020; Wang et al. 2021; Zhang, Zhao, and Yang 2019), and topographic indices regulate solar radiation and heat redistribution (slope, aspect, and elevation) (Guo, Hu, and Schlink 2022; Liang et al. 2023; Wang, Luo, and Peng 2020; Zhang, Zhao, and Yang 2019). In addition to these widely used variables, several other parameters have also been demonstrated to provide complementary information for LST downscaling. For example, soil indices (e.g. SMI, EMBI) (Sultana and Satyanarayana 2020; Sun, Zhou, and Liu 2019), surface albedo, and tasseled cap transformation (TCT) components (brightness, greenness, and wetness) (Li et al. 2024), which integrate multi-band spectral information into physically interpretable indicators. However, over-reliance on domain-specific expertise for feature variable selection results in incomplete exploration of the complex inter-variable relationships. Furthermore, underutilization of multi-source data synergies and inadequate optimization strategies for high-dimensional data spaces constrain the model performance. Feature interaction optimization techniques can effectively capture and utilize the complex interactions between variables, as reported by Chen and Luo (2024). A multi-level feature interaction network can reduce boundary blurring in remote sensing images. Li et al. (2022) introduced multi-order interaction methods for hyperspectral image classification, significantly improving feature discriminability. The incorporation of feature interactions addresses the issue of neglecting variable interaction effects in traditional methods. Shapley additive explanation (SHAP) provides quantifiable feature importance metrics (Mansourmoghaddam et al. 2024; Yu et al. 2020), which can avoid two limitations of conventional methods, namely: the black-box nature of feature selection and the inability to assess interaction effects.

In LST downscaling studies, the construction and selection of machine learning models necessitate a strategic equilibrium between model applicability and computational complexity (Chen et al. 2014; Xie and Li 2025; Zhu, Zhou, and Krisp 2025). From simple multiple linear regression to complex ensemble learning models, the performance of different models varied across scenarios (Jaber and Sengupta 2024; Li et al. 2022). For instance, Wu and Li (2019) achieved commendable downscaling results with a random forest (RF) regression model, but its ability to capture higher-order variable interactions may compromise predictive efficacy in heterogeneous surface environments. Conversely, deep learning models offer significant advantages in handling nonlinear relationships; their limitations stem from high computational complexity and inherent "black-box" nature. Consequently, selecting the most suitable machine learning model remains a critical issue, requiring a balance between model predictive performance, computational efficiency, and result interpretability.

To address the aforementioned challenges, this study proposes an innovative machine learning-based LST downscaling framework integrating feature interaction optimization with SHAP scoring. The approach aims to capture the complex inter-variable relationships through feature interaction and SHAP scores, thereby enhancing the model's ability to process high-dimensional data and nonlinear relationships, and ultimately improving the accuracy and operational stability of the LST downscaling model.

A comprehensive comparison analysis was performed among various machine learning models, including linear models (such as ridge regression and elastic net), ensemble learning models (such as XGBoost and LightGBM), and spatial statistical methods such as spatial random forest (SRF). The investigation aims to identify the strengths and weaknesses of different models in the context of an interaction feature space, explore how optimal model selection can enhance downscaling accuracy, and ultimately select the best model for LST downscaling.

2. Materials

2.1. Study area

The Dahei River Basin (DHR) (39°56'38"N–41°26'38" N, 110°20' E–112°50'43" E) is located in the Inner Mongolia Autonomous Region of China (Figure 1). As a vital tributary of the Yellow River, the Dahei River originates in Zhuozi County and flows from the northeast to the southwest, passing through Zhuozi County, Saihan District, Yiquan District, Tumote Zuoqi, and Toko County in Hohhot City, Inner Mongolia. Characterized by a typical continental monsoon climate (Wu et al. 2023), the DHR has a mean annual temperature of 6.2°C, and a mean annual precipitation of 394.7 mm. The basin demonstrates significant altitudinal variation, with the elevation of the headwaters at 1686.5 m decreasing to 986.7 m at the river mouth. The predominant vegetation in DHR is herbaceous plants, which cover more than 90% the regional area.

2.2. Data

2.2.1. Satellite data

Landsat 9 is equipped with two primary sensors that enable multispectral observations spanning from the visible to the shortwave infrared spectral ranges: the Operational Land Imager-2 (OLI-2) and the Thermal Infrared Sensor-2 (TIRS-2). The OLI-2 sensor provides nine spectral bands, including a 15-m resolution panchromatic band and eight 30-m resolution multispectral bands, which cover a spectral range from 0.433 μm to 2.300 μm. The TIRS-2 sensor provides two thermal infrared bands at a 100-m spatial resolution, specifically used for surface temperature monitoring (Akhyar et al. 2024). The TIRS-2 instrument is an upgraded version of its Landsat-8 predecessor, enhancing reliability by correcting stray light issues (Masek et al. 2020). In this study, Landsat 9 TIR data

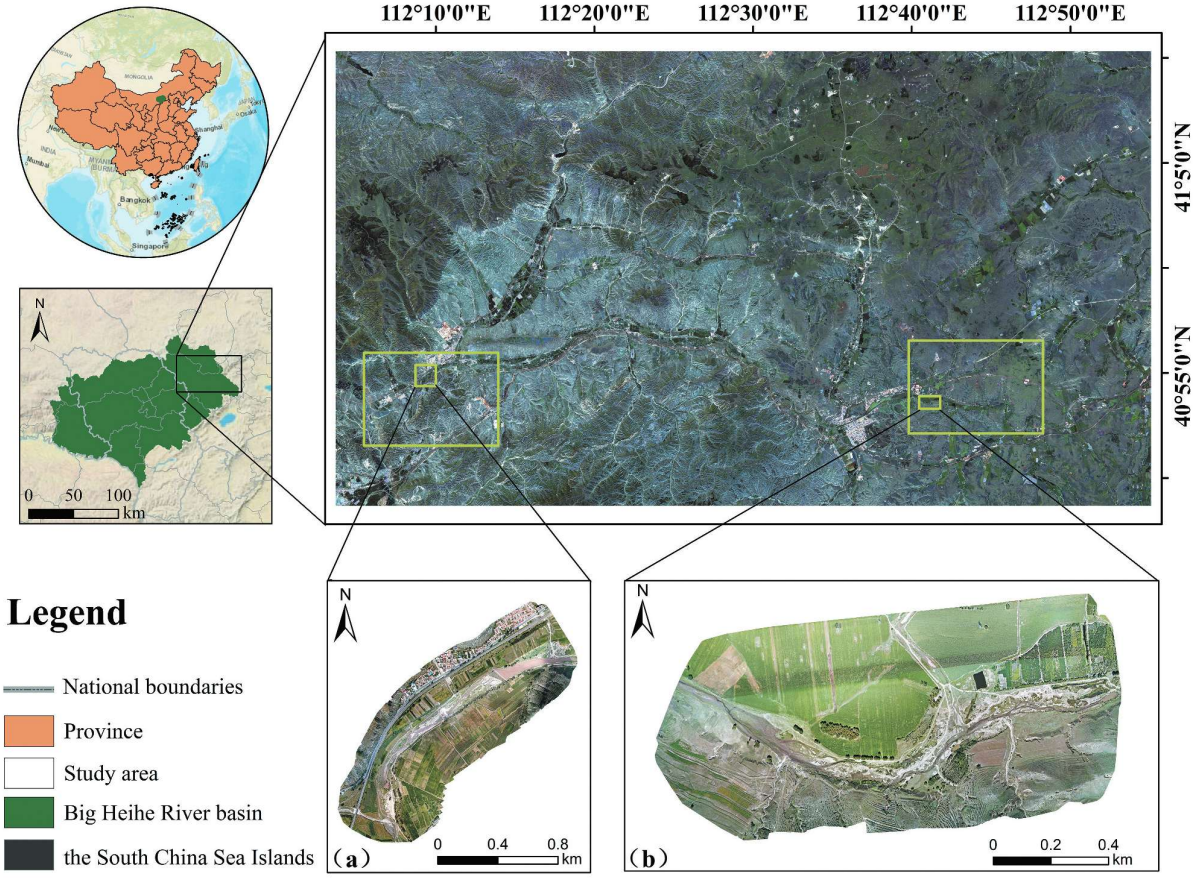


Figure 1. The location and unmanned aerial vehicle (UAV) image of the study area. (a) is Qixiaying; (b) is Zhuozishan. China's ministry of Natural Resources, China's standard administrative Division data GS (2024) 0650.

on 27 August 2024, over the study area were downloaded from Google Earth Engine (GEE), with rigorous quality control to exclude cloud-contaminated images. Landsat 9 operates in a sun-synchronous orbit with a 16-d revisit cycle, which, when combined with Landsat-8, achieves global coverage every 8 d. Five spectral bands, red, green, blue, near-infrared (NIR), and short-wave infrared (SWIR), along with derived LST, were used in this study.

The Sentinel-2 mission comprises twin satellites, Sentinel-2A and Sentinel-2B, equipped with high-resolution multispectral imaging capabilities. Each satellite independently achieves a 10-d revisit cycle at the Earth's equator. When the two satellites operate together, it enhances temporal resolution to 5 d (Claverie et al. 2018). These satellites carry multispectral instruments that capture images across 13 spectral bands spanning visible (VIS), NIR, and SWIR, with spatial resolutions varying from 10 m to 60 m. In this study, five spectral bands (red, green, blue, NIR, SWIR) from Sentinel-2 Level-2A atmospheric corrected surface reflectance images were used for experimental analysis.

To reduce errors in temperature observations caused by temporal mismatches, the Landsat 9 and Sentinel-2 images were acquired on 27 August 2024, 11:26:42 (Beijing Time, UTC +8). This acquisition date represents the closest available temporal match to the drone-collected thermal infrared data obtained during August 23–24, 2024.

2.2.2. Topographic data

Topographic data were from the Shuttle Radar Topography Mission version 3.0 (SRTM V3.0 available at https://srtm.csi.cgiar.org/srtmdata), with elevation, slope, and aspect data derived through the GEE platform at a 30-m spatial resolution. Through resampling, the processed terrain data were subsequently employed as a regression kernel in machine learning-based downscaling of the Landsat 9 TIRS-2 data (100 m resolution) and the Sentinel-2 data (10 m resolution).

2.2.3. Observed data

To validate the model, thermal infrared data were obtained from two distinct geographical locations using UAV platforms: the Qixiaying (40°57'31" N–40° 58'13" N, 112°7'4" E–112°8' E) on 24 August 2024, at 13:46:41 (Beijing time) (Figure 1(a)); and the Zhuozishan (40°54'23" N–40°54'48" N, 112°38'30" E–112°39'31" E) on 23 August 2024, at 12:24:41 (Beijing time) (Figure 1(b)).

Thermal infrared data were acquired via UAV in the study area on August 23–24, 2024, using a DJI H20T quad-sensor payload. The UAV was operated at an altitude of 512 m above ground level (AGL), achieving a mean ground sampling distance (GSD) of less than 1 m. The GSD, representing the spatial scale of a single pixel on the ground (i.e. each pixel corresponds to an area smaller than 1 m × 1 m), is determined by sensor parameters (e.g. focal length and detector resolution) and flight altitude. This spatial resolution (GSD < 1 m) enables high-precision land surface temperature mapping. All UAV-derived data were processed in Pix4D Mapper photogrammetric software.

3. Methods

3.1. Procedure

An integrated methodological framework for multi-scale land surface temperature retrieval was proposed, which combines multi-source remote sensing data fusion and machine learning modeling to enhance the precision and reliability of high-resolution (10 m) thermal mapping. The workflow includes Multi-sensor data preprocessing, variable selection, model development, and high-resolution LST estimation (Figure 2).

Step 1. Multi-sensor data preprocessing: The workflow begins with integrating optical bands, remote sensing indices, and topographic features from Landsat 9 at 100 m resolution and Sentinel-2 at 10 m resolution (Table 1). RGB bands, including the red, green, and blue bands, are chosen as single-band variables. The red band demonstrates strong differentiation between vegetation and non-vegetation areas. Vegetation exhibits lower reflectance (due to chlorophyll absorption), correlating with lower surface temperatures (Moshou et al. 2011), while bare soil and buildings show higher reflectance and elevated surface temperatures (Koroleva et al. 2017; Yang et al. 2010). The green band captures the peak vegetation reflection band and is particularly sensitive to the physiological state of vegetation. Healthy vegetation has higher green-band reflectance, typically associated with cooler surfaces (Broge and Leblanc 2001; Zhang et al. 2008). The blue band can be used for building shadow detection, providing insights into micro-environmental thermal variation (Lindberg and Grimmond 2011; Ok 2013). Remote sensing indices include enhanced vegetation index (EVI), normalized difference built-up index (NDBI), and normalized difference moisture index (NDMI). EVI quantifies vegetation cover and health, and offers improved sensitivity in high-biomass regions by incorporating the blue band to mitigate atmospheric and soil background interference (Huete et al. 2002; Peng et al. 2018). NDBI identifies urban buildings and artificial surfaces by the reflectance difference between the SWIR and NIR bands (Estoque and Murayama 2015; Ghosh, Kumar, and Kumari 2022); therefore, it shows more stable correlation with surface temperature and minimal seasonal variability (Pandey et al. 2024).

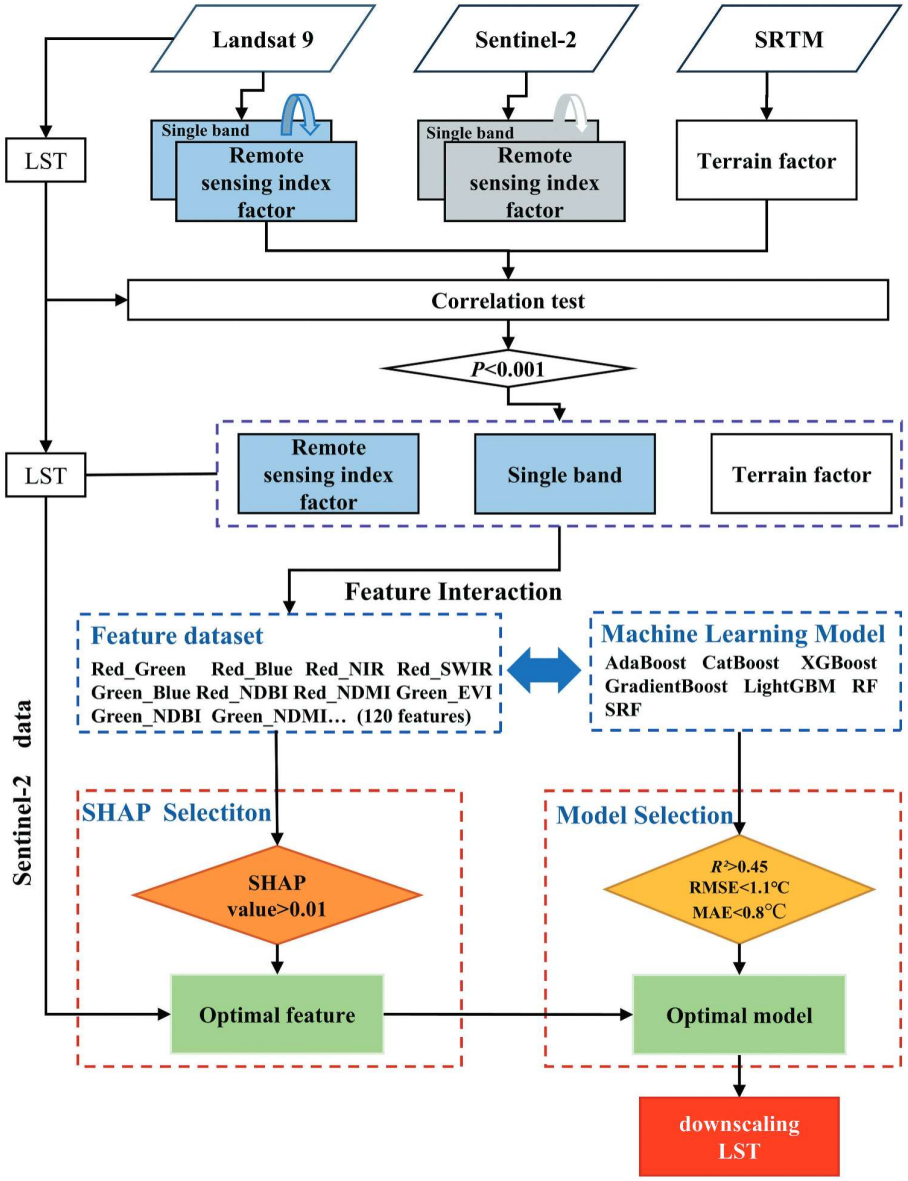


Figure 2. Flowchart of proposed methodology for LST downscaling.

Table 1. Remote sensing variables.

Feature name	Landsat 9	Sentinel-2
Red	Band 4 (640–670 nm)	Band 4 (664 nm)
Green	Band 3 (530–590 nm)	Band 3 (560 nm)
Blue	Band 2 (450–510 nm)	Band 2 (496 nm)
NIR	Band 5 (850–880 nm)	Band 8 (835 nm)
SWIR	Band 6 (1570–1650 nm)	Band 11 (1613 nm)
EVI	$2.5 \times ((B_5 - B_4)/(B_5 + 6 \times B_4 - 7.5 \times B_2 + 1)$	$2.5 \times ((B_5 - B_4)/(B_5 + 6 \times B_4 - 7.5 \times B_2 + 1)$
NDBI	$(B_6 - B_5)/(B_6 + B_5)$	$(B_{11} - B_8)/(B_{11} + B_8)$
NDMI	$(B_5 - B_6)/(B_5 + B_6)$	$(B_8 - B_{11})/(B_8 + B_{11})$

NDMI assesses vegetation moisture content via NIR-SWIR reflectance difference (Nicoletti et al. 2024). Generally, stronger moisture stress correlates with elevated surface temperature (Viswambharan, Kumaramkandath, and Tali 2022). The selected indices comprehensively reflect surface feature information. EVI reflects vegetation transpiration and cooling effects, NDBI captures urban heat retention dynamics, and NDMI indicates moisture stress levels and transpiration potential. The SRTM topographic features are resampled to 100 m (Landsat 9) and 10 m (Sentinel-2) resolution, ensuring spatial consistency with optical bands and remote sensing indices.

Step 2. Variable selection: Through correlation analysis and feature interaction design, the nonlinear relationships between variables are systematically

explored, and the high-contribution features are selected using the SHAP method as model inputs. The predictive models directly incorporate the 11 baseline variables into three linear regression frameworks: (1) ordinary least squares (OLS) as a reference, (2) ridge regression to penalize correlated variables via L2 regularization, and (3) elastic net combining L1/L2 penalties for automated feature selection. This multi-method design inherently mitigates multicollinearity risks while preserving the interpretability of raw variables. The variable selection in the machine learning model has three steps: correlation calculation, feature interaction, and SHAP values evaluation. First, the Spearman correlation between all satellite bands, vegetation indices, terrain indices, and LST are calculated. We selected variables significantly correlated with LST at $p < 0.001$ as predictors.

Regarding the calculation of feature interaction quantity, further clarification is required. When constructing feature interactions, we use combinations rather than permutations. This is because the interaction between features is bidirectional. For example, the interaction between the red and green bands is mathematically the same (red × green = green × red). Therefore, although it appears to be pairwise interactions among 11 variables, the correct calculation method should be based on combinations without repetition in combinatorial mathematics. We classify the features into four categories: bands, indices, terrain features, and non-linear transformations. The interactions within and between these categories are then computed, yielding 105 unique interaction features. In total, 120 variables (including original predictors and interaction terms) were used for model training, and the complete list of these variables is provided in the supplementary materials (Appendix B). This calculation method not only avoids repeated statistics of features but also aligns more closely with the actual physical meaning and mathematical principles of feature engineering. By constructing the feature interaction system in this way, we can more effectively capture the non-linear relationships between variables, thereby enhancing the model's expressive power.

In feature engineering, we directly use the raw data for interactions (multiplication) rather than performing normalization. This preserves the scale effects of the original data. Magnitude differences between certain variables, such as terrain and spectral data, may carry important geographical information, and normalization would eliminate these valuable scale characteristics. Additionally, this method preserves the physical meaning of the data, ensuring that the feature interaction results have clear interpretability, avoiding potential changes in distribution characteristics that may arise from the normalization process. Finally, the SHAP method is applied to quantify feature contribution, and the variables with high contribution rates were used as the regression core. SHAP values provide an explanation for the output of the machine learning model. In machine learning, each feature is assigned an importance value, namely the SHAP value, that indicates the influence magnitude on individual predictions and pairwise importance rankings among features (Aldrees et al. 2024). SHAP values are model-agnostic; they can be used in both linear regression and decision trees (Zhang, Wang, et al. 2024).

Step 3. Model development: Regression kernels are developed using both linear and machine learning models. The linear model was developed on the 11 basic features, while the machine learning model predicts Landsat 9 LST using SHAP-optimized features in step 2 through a 100 m resolution training framework. To assess the effectiveness of the machine learning model, 10-fold cross-validation is implemented to ensure model stability and generalization. Each cross-validation employs a tripartite evaluation protocol: the coefficient of determination (R^2) quantifies the proportion of LST variability captured by predictive features, the root mean square error (RMSE) measures the deviation between predicted and observed temperature, and the mean absolute error (MAE) provides an intuitive measure of prediction errors. Model acceptability thresholds are defined based on the minimal R^2 value observed across all candidate models in the training dataset, ensuring the threshold reflects actual modeling performance boundaries ($R^2 > 0.45$, RMSE < 1.11°C, and MAE < 0.8°C). Models that meet all three criteria are considered valid and suitable for subsequent surface temperature estimation.

Step 4. High-resolution LST estimation: The SHAP-optimized features (step 2) derived from Sentinel-2 data, integrated with SRTM topographic features at 30-m resolution, were applied in the validated model (step 3) to generate LST data at 10-m resolution.

3.2. Method assessment

3.2.1. UAV-based validation

The downscaled LST was validated using UAV-derived thermal observations with a spatial resolution of < 1 m (GSD). To align the UAV thermal data with the downscaled LST product (10 m resolution), the original UAV data were resampled to 10 m resolution using an averaging aggregation method (Chen et al. 2023), which minimizes upscaling errors in heterogeneous landscapes by preserving thermal characteristics across scales. To ensure representative validation, a stratified random sampling strategy was adopted, focusing on homogeneous land cover units to reduce spatial heterogeneity impacts (Wang, Haining, and Cao 2010). Six validation sites (3 vegetated, 3 bare soil) were selected based on the following criteria:

(1) Spatial homogeneity: Each site exhibited low intra-site variability in land cover, confirmed by thermal spatial uniformity (LST standard deviation < 6.4°C).
(2) Land cover representativeness: Sites were chosen to reflect dominant land cover types within the study area, avoiding mixed-pixel zones.
(3) Geographic distribution: Sites were evenly distributed across the study region to capture spatial variability.

Each validation unit comprised a contiguous 10 × 10 pixel matrix (1 hectare at 10-m resolution). This approach ensured statistically robust comparisons while aligning with the spatial resolution of the downscaled LST product.

3.2.2. Model evaluation index

To ensure both scientific rigor and generalizability, this study selected a rectangular region covering diverse land-use types, including mountains, vegetation, built-up areas, and water bodies. The region comprises 239 rows × 192 columns of 100 m resolution pixels, totaling 45,888 samples used as the training set. This dataset size not only guarantees the model's adaptability across different surface conditions but also avoids the instability that may arise from an insufficient number of samples. Three statistical metrics were employed to quantify the predictive performance of the machine learning models: R^2 (Equation (1)), RMSE (Equation (2)), and MAE (Equation (3)). R^2 quantifies the model's explained variance to LST spatial heterogeneity, RMSE represents the deviation between predicted and actual values, and MAE provides an absolute error measurement.

These metrics are primarily used for:

(1) Model selection at 100-m scale. Comparative evaluation of candidate machine learning models on the training dataset (100-m resolution) to select the optimal downscaling model configuration and parameter combination.
(2) Generalization ability assessment: Quantitative analysis using an independent 10-m resolution validation set to evaluate the model's generalization ability and detect overfitting.
(3) Fine-Scale Accuracy Verification: Direct comparison with UAV-derived thermal infrared data (10 m aggregated data) to assess the model's downscaling performance at finer scales.
(4) Comparison between downscaled LST (10 m) and corresponding Landsat 9 LST raster (100 m). Given that grassland (77.6%), cropland (13.5%), forest (7.6%), and impervious surfaces (1.1%) account for over 99% of the study area. We randomly selected 100 points from each of the four land types to compare the downscaled effect.

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (\hat{y}_i - y_i)^2}{\sum_{i=1}^{n} (\bar{y} - y_i)^2} \quad (1)$$

$$\mathrm{MAE} = \frac{1}{n} \sum_{i=1}^{n} |\hat{y}_i - y_i| \quad (2)$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2} \quad (3)$$

where $\hat{y}$ is the model prediction, y_i the observed value, and n is the number of samples in the training or validation dataset.

3.3. Methodology implementation

3.3.1. Correlation analysis between candidate variables

Spearman correlation analysis between 11 fundamental variables, including single-band spectral features (red, green, blue, NIR, SWIR), spectral remote sensing indices (EVI, NDBI, NDMI), terrain indices (elevation, slope, aspect), and LST, is shown in Figure 3. The single-band spectral features exhibited the strongest correlation with LST, with the maximum correlation coefficient occurring in the red band ($r = 0.62$), followed by the blue band ($r = 0.61$), SWIR ($r = 0.39$), and NIR ($r = -0.31$). The red and blue bands demonstrated the strongest positive correlations with LST, suggesting that surface areas exhibiting enhanced reflectance in these spectral regions correspond to elevated thermal emissions.

The spectral indices showed moderate correlations with LST: NDBI ($r = 0.53$, $p < 0.001$), NDMI ($r = -0.53$, $p < 0.001$), and EVI ($r = -0.51$, $p < 0.001$). Notably, NDBI displayed a significant positive correlation with LST, indicating that increased building density contributes to elevated surface temperatures via the urban heat island effect. Conversely, NDMI revealed a strong negative correlation with LST, suggesting that temperature increases are associated with decreased vegetation moisture. Similarly, EVI demonstrated a negative correlation, highlighting vegetation cover's critical role in temperature mitigation through evapotranspiration and shading effects. These patterns reflect distinct relationships between land surface characteristics and LST, providing insights for environmental monitoring.

Among the terrain indices (Figure 4), the slope showed a weak positive correlation ($r = 0.14$), while aspect and elevation displayed negative correlations of $r = -0.12$ and $r = -0.03$, respectively. The terrain

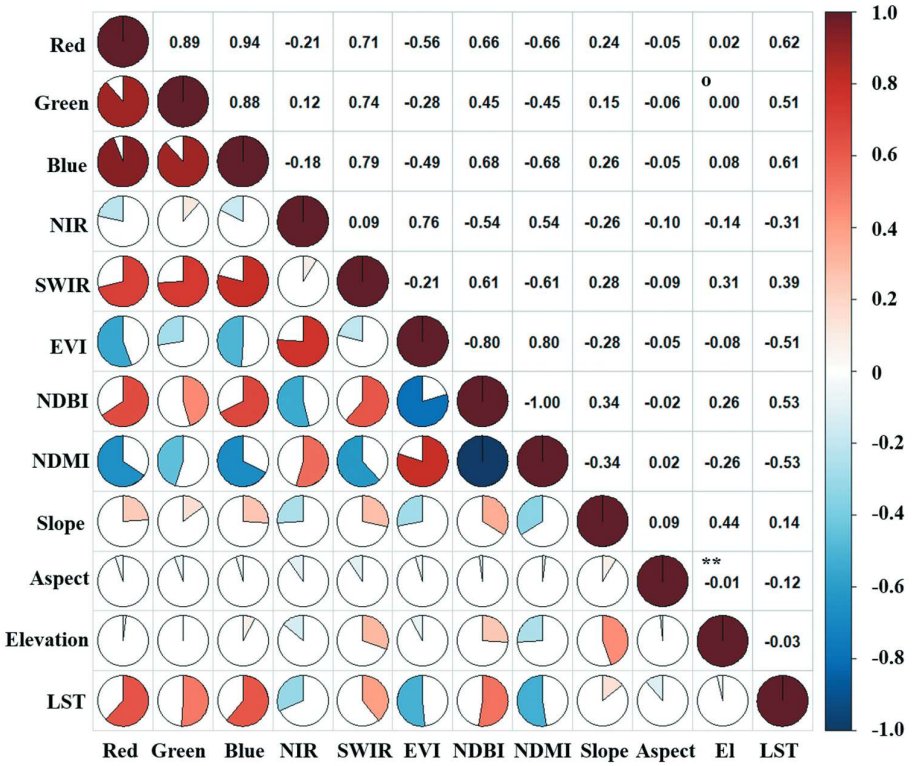


Figure 3. Correlations between the candidate features and LST. El is the elevation, ** is significant at 0.01, o is not significant, and the rest are significant at 0.001.

factors demonstrated the limited statistical associations with LST ($|r| < 0.15$), suggesting minimal influence on surface temperatures in the study area.

The inter-variable correlation patterns reflect urbanization-driven impacts on surface ecohydrological processes. Rapid urban expansion, characterized by the replacement of vegetated pervious surfaces with impervious materials (e.g. concrete, asphalt), directly inhibits evapotranspiration and soil moisture retention. This anthropogenic hydroclimatic alteration manifests as a strong negative correlation between NDBI and NDMI ($r = -1.0$), indicating that intensified built-up areas spatially coincide with depleted vegetation moisture. Furthermore, visible spectrum bands (red, green, blue) exhibit strong inter-band correlations ($r > 0.88$), attesting to the inherent spectral consistency of the dataset.

3.3.2. *Feature importance analysis with SHAP*

In the machine learning approach, SHAP values were computed to quantify the relative contribution and predictive importance of each feature in LST prediction. Features with low contributions were eliminated while retaining those significant variables demonstrating substantial influence on model outputs. The average

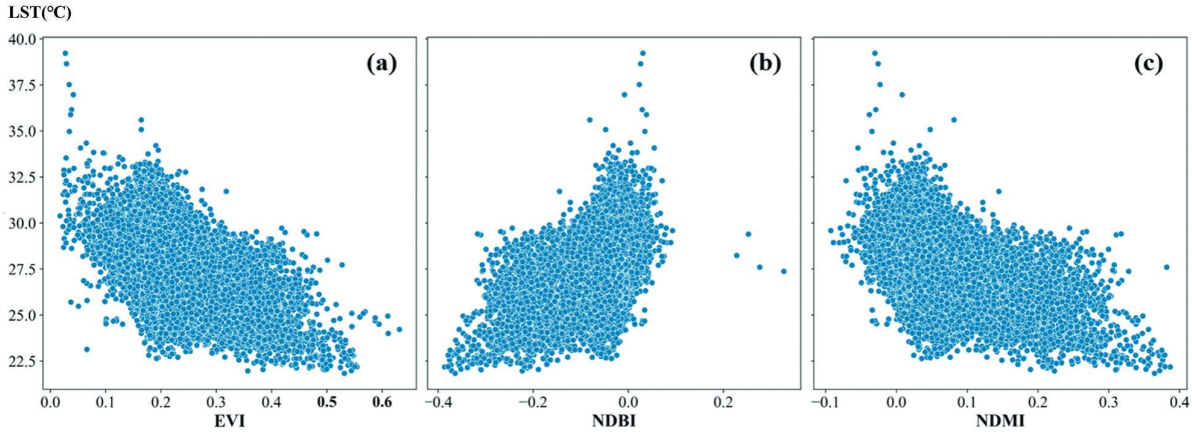


Figure 4. Correlation between the remote sensing index and LST.

absolute SHAP values of the top 10 features are shown in Figure 5. The red-blue spectral interaction variable exhibited the highest stability and criticality across five model performances, showing particular prominence as it was consistently identified as the most important feature in LST downscaling models. The red band also showed significant importance across multiple models, especially in AdaBoost with a feature importance score of 0.3696, followed by RF (0.1609) and SRF (0.1452). It also performed well in tree-based gradient boosting models, achieving importance values of 0.1430 in CatBoost, 0.1383 in LightGBM, and 0.1037 in GradientBoost. Remote sensing index-derived variables (e.g. EVI_Elevation and EVI_Aspect) ranked among the top contributors in multiple models, with EVI_Elevation attaining peak scores in CatBoost (0.1692) and GradientBoost (0.1862), indicating their high explanatory power when integrated with terrain features. In contrast, terrain interaction variables exhibited model-specific dominance: Elevation showed significant predictive power in tree-based gradient boosting models, with notable feature importance scores in XGBoost (FI = 0.1951) and CatBoost (FI = 0.0681).

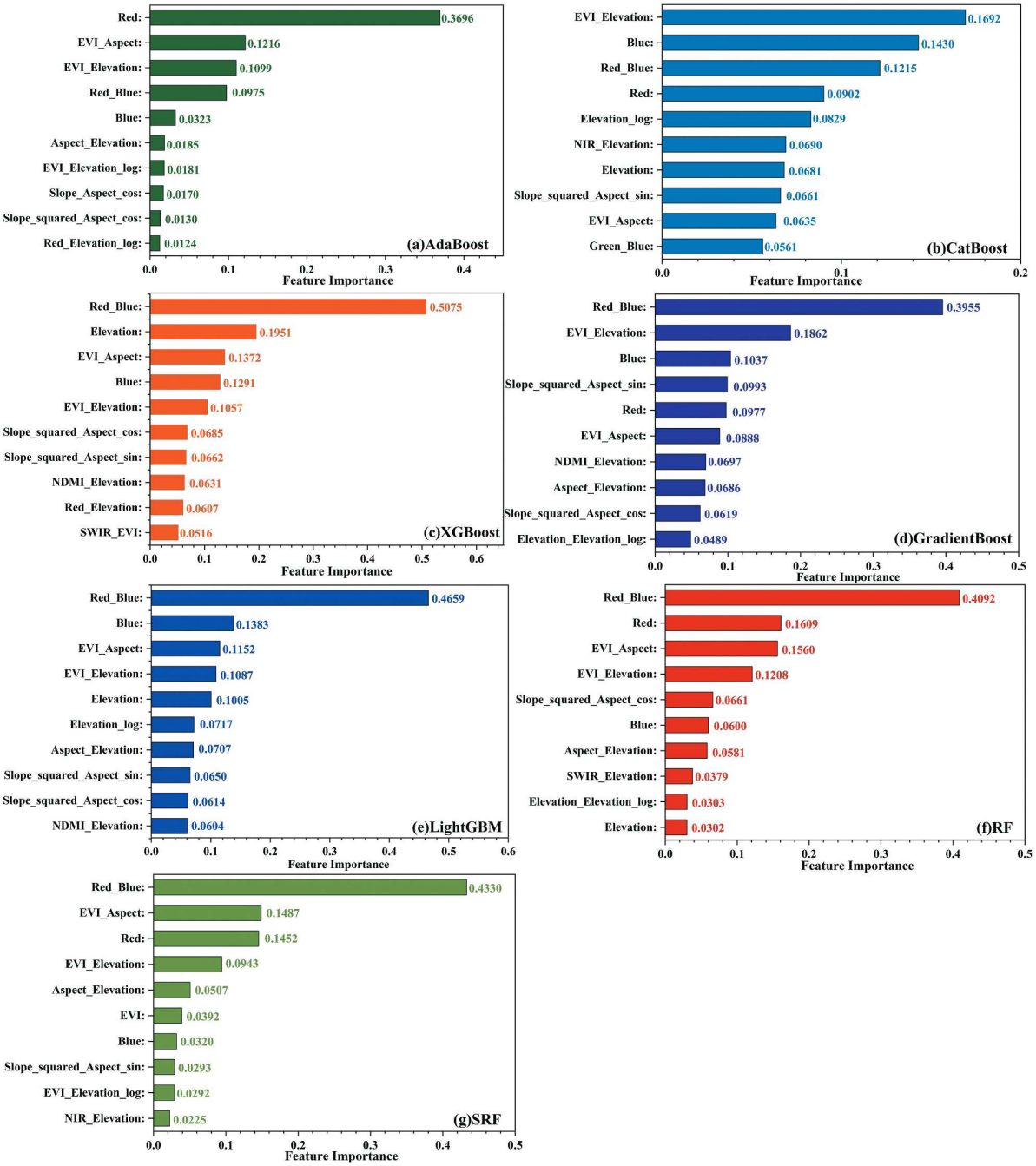


Figure 5. Characteristic importance analysis with the SHAP model.

The slope interaction variables (e.g. Slope_squared_Aspect_cos and Slope_squared_Aspect _sin) ranked among the top 10 most important features in both XGBoost and GradientBoost models. Notably, the contribution of Slope_squared_Aspect_sin was particularly significant (feature importance score = 0.0993) in the GradientBoost model. The cumulative contribution of the top 10 features captured 85–90% of the total predictive variance (Figure 5). Despite the maintained statistical significance of lower-ranked features (SHAP value > 0.01), their limited practical utility motivated the implementation of a core predictor cohort.

The influence magnitude and direction of various features on surface temperature prediction are shown in Figure 6. In the AdaBoost model, the red band was most prominent (SHAP values ranging from −1.0 to 0.4), followed by EVI_Aspect and EVI_Elevation. CatBoost was predominantly influenced by EVI_Elevation (SHAP values ranging from −0.6 to 0.6), with blue and red-blue features following closely. The remaining five models (XGBoost, GradientBoost, LightGBM, RF, and SRF) consistently prioritized the red–blue interaction feature, showing the widest SHAP value range (−2.0 to 2.0). Among these, XGBoost and RF exhibited the most pronounced variability in SHAP values. LightGBM and SRF also showed significant contributions from the blue band. Most influential features displayed asymmetric SHAP value distributions, suggesting nonlinear associations with surface temperature. Terrain interaction features (e.g. Slope_squared_Aspect_sin) had relatively concentrated SHAP value distributions, indicating stable and consistent effects on model predictions.

3.3.3. Downscaling model construction

This study aims to identify the key hyperparameters of machine learning-based downscaling models to enhance their performance. Specifically, the focus is on two fundamental hyperparameters commonly used in tree-based models, n_estimators and max_depth, both of which play a crucial role in determining model complexity and learning capacity. The n_estimators parameter determines the number of trees in the ensemble, thereby influencing the model's capacity to fit training data and its computational cost. In contrast, max_depth specifies the maximum depth of each tree, which directly affects the model's complexity and its susceptibility to overfitting (Shi et al. 2024).

We employ 100 m LST data along with a comprehensive set of predictor variables as model inputs. During the training process, the predicted LST values are evaluated against the observed 100 m LST data to assess model accuracy. To ensure robust hyperparameter optimization and minimize overfitting, we employed a 10-fold cross-validation strategy combined with grid search, computing the R^2 for each hyperparameter combination (Table 2).

SRF achieved the best performance in LST downscaling, with an R^2 of 0.9594, and RMSE and MAE values of 0.3151 and 0.2418, respectively (Table 3). Traditional RF exhibited similar predictive performance, with an R^2 of 0.9535. XGBoost and CatBoost also demonstrated reasonable predictive capabilities, with R^2 values of 0.7876 and 0.7490, respectively. In contrast, traditional linear regression models showed limited effectiveness, with ElasticNet performing the worst ($R^2 = 0.2711$). Linear models exhibited a consistent pattern with the original LST but failed to capture detailed thermal variations in heterogeneous landscapes. Among the machine learning models, AdaBoost demonstrated the weakest performance (Table 3). In contrast, the other ensemble methods produced more stable and reliable results, with RF and SRF delivering the best outcomes (Figure 7).

Figure 8 presents a comparative analysis of 10 distinct models for LST downscaling. The correlation coefficients range from 0.71 to 0.98, while the standard deviations (SD) span from 0.40 to 1.41. The SRF and RF models demonstrated superior performance, with their data points positioned close to the reference line (Ref). Among them, the SRF model achieved the best performance, with a correlation coefficient of 0.9827 and an SD of 1.4115—the closest to the reference SD of 1.5633—highlighting its strong ability to capture LST spatiotemporal variability.

In contrast, conventional linear models (Linear regression and Ridge) showed limited predictive performance, with correlation coefficients around 0.77 and SD values near 1.2. The ElasticNet model exhibited the poorest performance, with a low correlation coefficient of 0.6499 and a substantially reduced SD (0.4081), suggesting an underestimation of LST variability.

Ensemble learning models, including XGBoost, CatBoost, and LightGBM, showed intermediate performance, with correlation coefficients ranging from 0.83 to 0.89. While these models exhibited moderate predictive capabilities, their accuracy remained inferior to that of the RF-based models. The robust performance of RF-based architectures in preserving LST spatiotemporal characteristics, as evidenced by near-perfect correlation and optimal SD alignment, supports their selection as the preferred framework for downscaling.

4. Results

4.1. Validation with UAV observation

The SRF model exhibited superior predictive accuracy in the vegetated sites, as evidenced by high coefficients of determination ($R^2 = 0.70$, 0.80, and 0.74) (Figure 9 (d–f)), indicating robust model explanatory capabilities in these regions. Although the MAE and RMSE values fluctuated, they remained within an acceptable range overall, suggesting well-controlled prediction

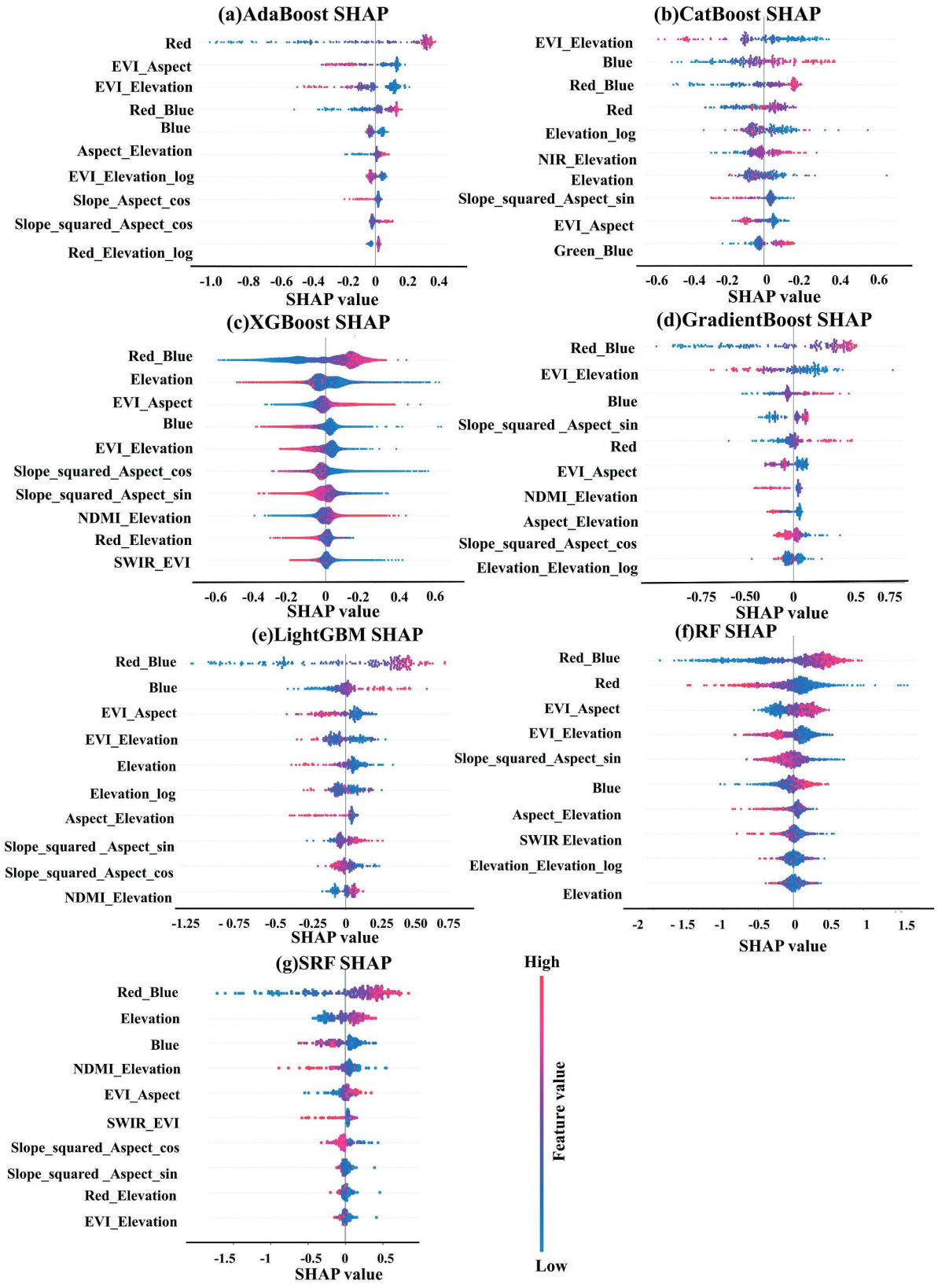


Figure 6. SHAP feature distribution. Each point represents a sample, the horizontal axis represents the SHAP value, and the color from blue to red indicates the feature value from low to high.

Table 2. Parameter selection of the tree model.

Name	AdaBoost	CatBoost	XGBoost	GradientBoost	LightGBM	RF	SRF
Number of trees	131	77	216	150	102	250	174
Maximum depth of trees	8	3	3	4	9	11	30
Minimum samples split	/	/	/	2	/	2	6
Minimum samples leaf	/	/	/	4	/	4	3
Maximum features	/	/	/	/	/	2	/
Bootstrap samples	/	/	/	/	/	FALSE	/
Learning rate	0.13	0.18	0.12	0.1	0.05	/	/
Subsample	/	/	0.7	0.8	0.9	/	/
Min child weight	/	/	2	/	/	/	/
Gamma	/	/	0.32	/	/	/	/
Reg alpha	/	/	0.89	/	0.5	/	/
Reg lambda	/	/	2.76	/	0.5	/	/
Colsample bytree	/	/	0.77	/	/	/	/
Min child samples	/	/	/	/	5	/	/
Min data in leaf	/	10	/	/	20	/	/

Table 3. Model evaluation on the training sets.

Model Name	R^2	RMSE	MAE
Linear regression	0.5929	0.9975	0.7731
Ridge	0.5788	1.0145	0.7889
ElasticNet	0.2711	1.3346	1.045
AdaBoost	0.4999	1.1056	0.8732
CatBoost	0.7490	0.7832	0.6098
XGBoost	0.7876	0.7205	0.5582
GradientBoost	0.6193	0.9645	0.7542
LightGBM	0.699	0.8577	0.6699
RF	0.9535	0.3371	0.2572
SRF	0.9594	0.3151	0.2418

errors. In the vegetated sites, SRF outperformed all comparative models across all evaluation metrics (R^2, RMSE, and MAE), with R^2 values exceeding 0.7 in selected sites (Figure 9(d–f)), demonstrating superior capability in capturing the spatial heterogeneity of LST. Notably, its RMSE and MAE values were significantly lower than those of other models, indicating minimized prediction errors and enhanced generalization capability. Linear models (e.g. Ridge and ElasticNet) showed higher R^2 ($R^2 = 0.91$) values but elevated RMSE and MAE (Figure 10), indicating severe overfitting. However, Tree-based models showed slightly lower R^2 values compared to SRF, suggesting a trade-off between error minimization and variance explanation (Figure 10), possibly due to over-smoothing of local nonlinear patterns. The model exhibited degraded performance in bare land areas, with sharply reduced coefficients of determination of 0.00, 0.26, and 0.17 (Figure 9(a–c)), indicating severely limited predictive capability. The regression line deviated significantly from the 1:1 line. Additionally, error metric analysis revealed higher MAE and RMSE values compared to the vegetated

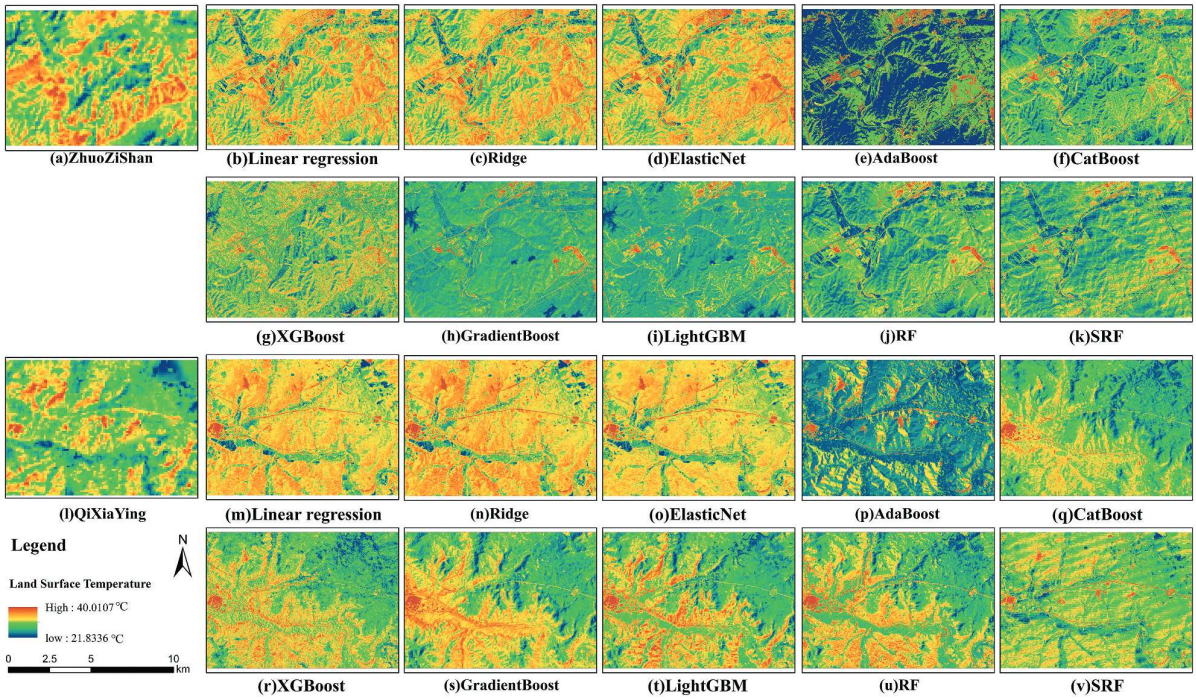


Figure 7. Spatial pattern of downscaled LST at 100 m resolution. (b)–(k) are downscaled LST in Qixiaying, (m)–(v) are downscaled LST in Zhuozishan, (a) and (l) are surface temperature images from Landsat 9.

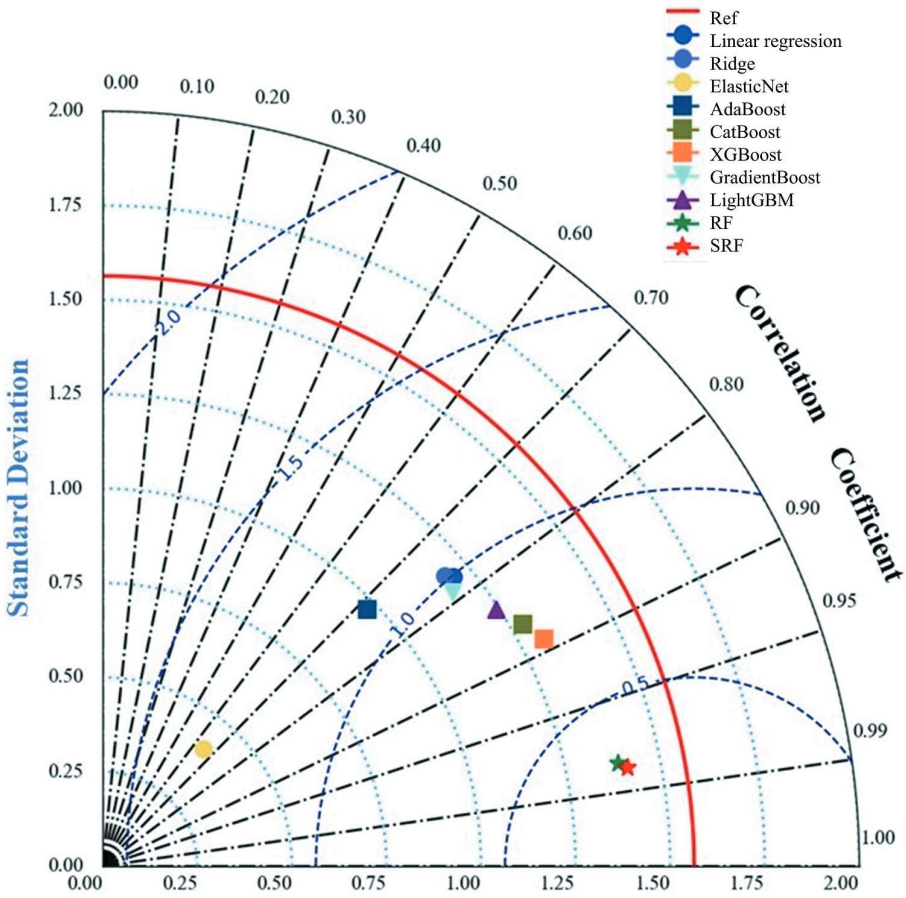


Figure 8. Model evaluation.

areas, reflecting a deterioration in predictive accuracy (The partial effects can be found in Appendix D).

4.2. Downscaled assessment

The spatial random forest exhibited outstanding performance in LST downscaling ($R^2 = 0.96$). These results (Figure 10, Table 3) indicate that the spatial random forest model, which explicitly incorporates spatial features, optimally balances the correlation strength and predictive variability of LST downscaling outputs.

The model performances in different land cover types are shown in Table 4. SRF and RF consistently outperformed in all four land use types. The error in farmland was relatively large among the four categories, and the linear model was difficult to fit, while the tree model was significantly superior. Forest had the lowest average RMSE and was easy to fit. Grassland SRF had a significant increase in gain compared to RF, indicating that there was still a significant interaction. The average error for Impervious was the largest among all land categories, and the differences between the models were the most obvious (Detailed parameters can be found in Appendix C).

The coarse delineation of land cover boundaries in linear method predictions (Figure 11) reveals two fundamental limitations: inability to resolve complex nonlinear decision boundaries, and poor discriminative performance when handling overlapping or nonlinearly separable spectral/spatial features across distinct land cover classes. Specifically, simple linear regression models fail to accurately differentiate target categories under these conditions, as evidenced by the misclassification artifacts (more detailed local enlargement diagrams are presented in Appendix D).

In contrast, ensemble models, which can capture nonlinear relationships, demonstrate superior capability in identifying complex land cover patterns, such as cropland and mountainous regions (Figure 11). The ensemble architectures exhibit enhanced robustness against noise while mitigating overfitting risks.

Compared to conventional methods, SRF and RF demonstrate superior capability in resolving thermal characteristic disparities across land cover types, particularly over homogeneous vegetated surfaces where spatial homogeneity dominates. Based on the first law of geography (Manning, Li, and Liu 2023), SRF incorporates an adaptive kernel-based spatial weight matrix (Chen, He, and Li 2024) that explicitly accounts for both distance decay effects and anisotropic

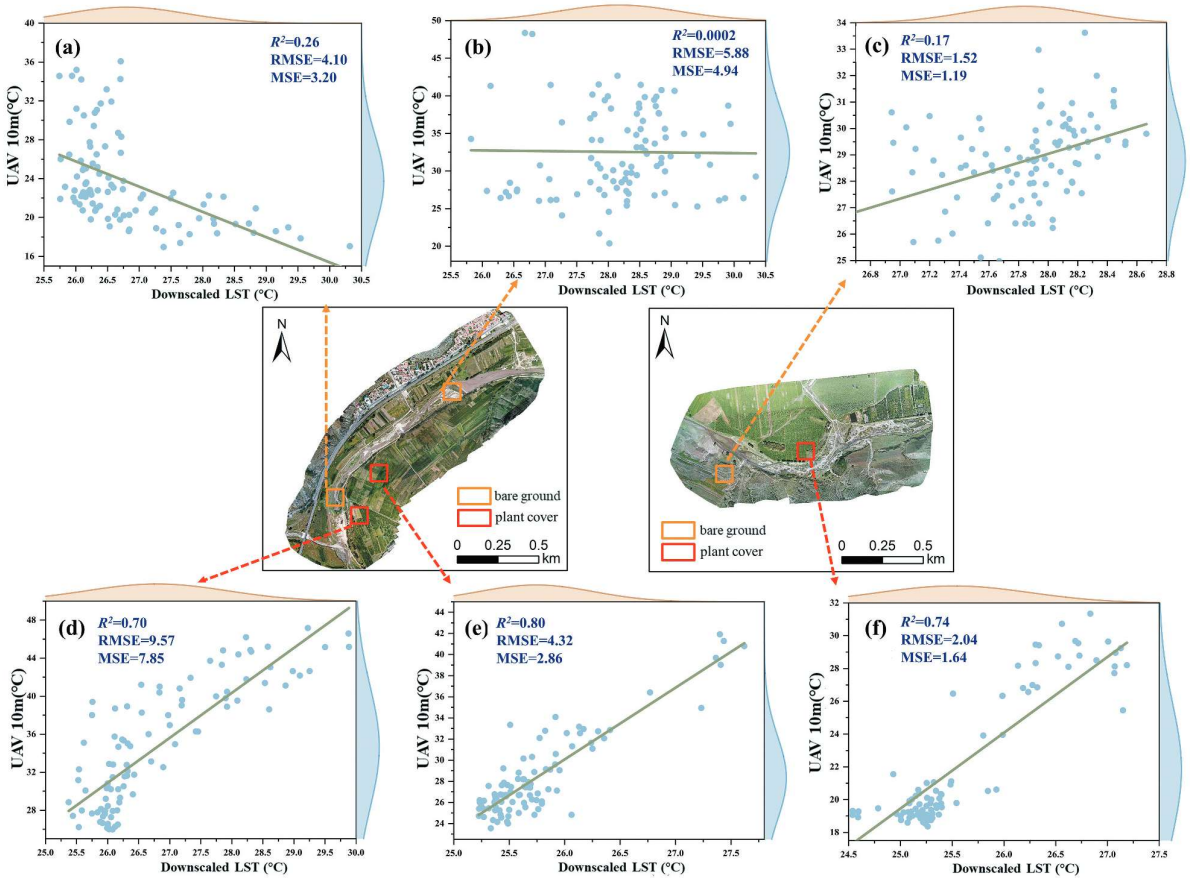


Figure 9. Validation of downscaled LST against UAV observations in different land cover types. *X*-axis is the downscaled LST (°C); *Y*-axis is UAV-observed surface temperature (°C). (a)–(c) are bare soil, (d)–(f) are vegetated surfaces.

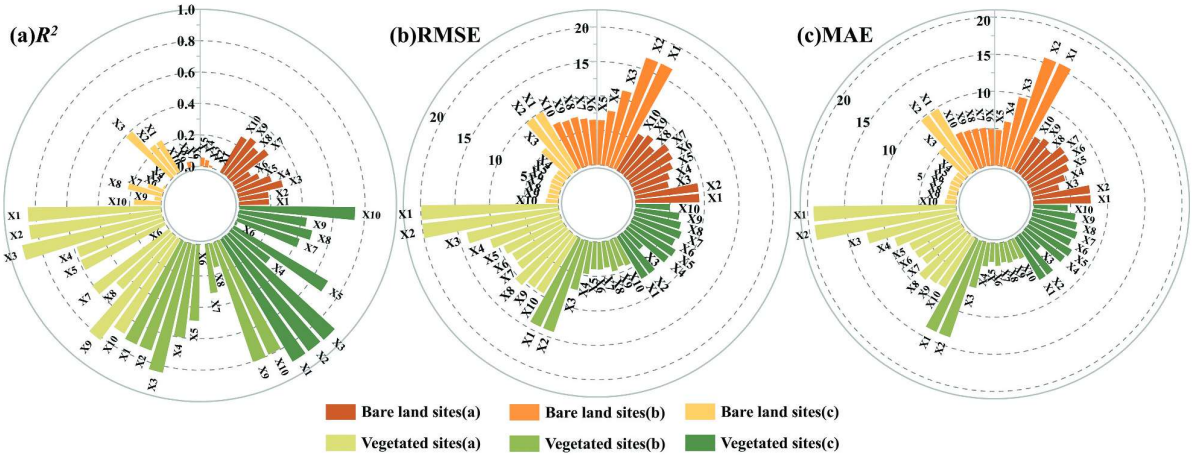


Figure 10. Comparison of evaluation results of different models. X1: linear regression; X2: Ridge; X3: ElasticNet; X4: Adaboost; X5: CatBoost; X6: XGBoost; X7: GradientBoost; X8: LightGBM; X9: RF; X10: SRF.

characteristics inherent in thermal fields. This weight matrix enables precise capture of the spatial dependency characteristics of LST's anisotropic distribution pattern arising from topography variations and spectra heterogeneity. Notably, in temperature gradient areas such as the building center–edge transition zone, SRF can accurately identify spatial discontinuities in the temperature field, a task that traditional machine learning methods (e.g. XGBoost, $R^2 = 0.79$) are unable to achieve. However, SRF overemphasizes the similarity of adjacent pixels, leading to excessive spatial smoothing and the formation of regular, wave-

Table 4. The R^2 between downscaled LST (10 m) and the corresponding Landsat 9 LST raster in different land cover type.

Model	Cropland	Forest	Grassland	Impervious
Linear regression	0.38	0.40	0.50	0.36
XGBoost	0.82	0.85	0.78	0.76
LightGBM	0.61	0.69	0.67	0.57
RF	0.91	0.92	0.90	0.91
SRF	0.94	0.93	0.97	0.97

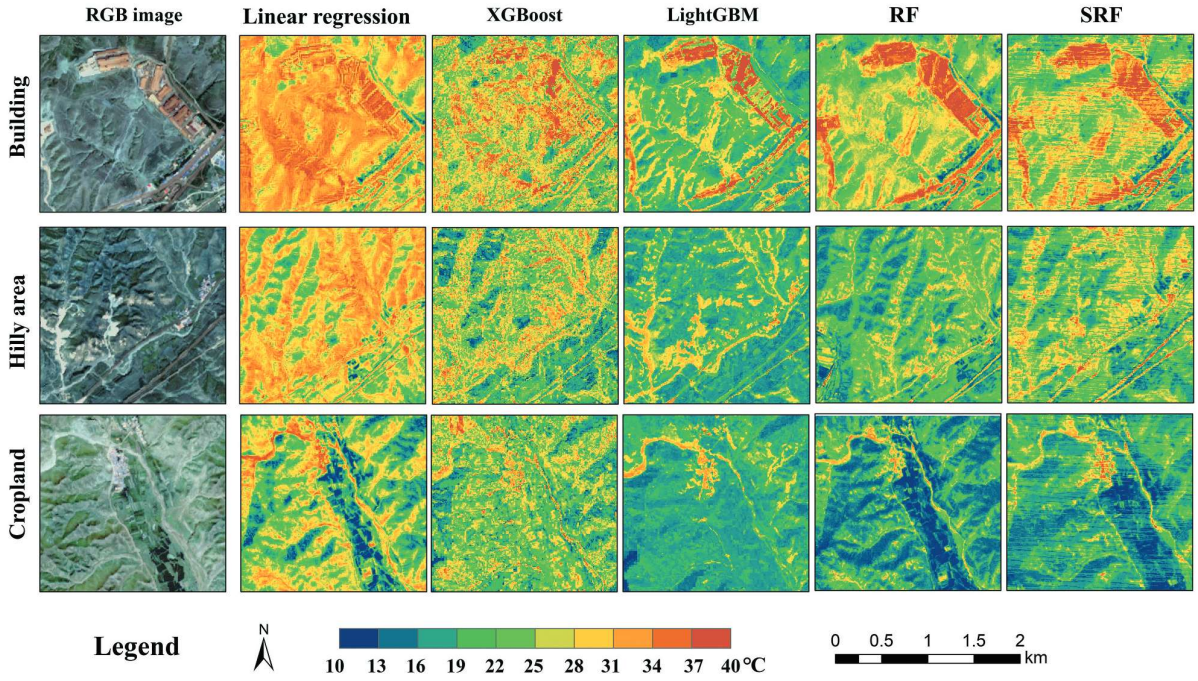


Figure 11. Downscaled LST in different land use types.

like strips rather than the true boundaries (Figure 11). This spatial over-smoothing phenomenon suggests that adjusting the spatial weight matrix parameters and optimizing kernel sizes are needed to improve classification accuracy.

The superior performance of SRF in LST downscaling ($R^2 = 0.96$) can be attributed to its unique spatial weighting mechanism. Unlike traditional random forest models that assume spatial independence between samples, SRF explicitly integrates spatial weights into the model construction process. By constructing spatially explicit decision trees, SRF accounts for the spatial autocorrelation present in geospatial data.

5. Discussion

5.1. Feature selection

In the cumulative SHAP value curves (Figure 12), the mean of the seven models shows that the cumulative SHAP contribution reaches approximately 1.0 when considering the top 10 highest-ranked features (sorted by SHAP value magnitude). This suggests that a small subset of input variables predominantly drives the model's predictive behavior. Such a pattern aligns with the sparsity of feature importance, where the majority of the model's output variability (over 80%) is captured by a limited number of highly influential features (Dev et al. 2022; Lundberg et al. 2020).

By selecting the top 10 features, we achieve two objectives: capturing the majority of the model's explanatory power and enhancing the interpretability and generalizability of the LST inversion framework. As the marginal increase in cumulative SHAP value rapidly diminishes beyond the tenth feature, including additional variables may introduce redundancy, increase computational cost, or even risk overfitting (Simon and Aliferis 2024; Theng and Bhoyar 2024). Such dimension reduction is essential for geospatial applications, as multi-source data fusion frequently generates high-dimensional input spaces.

Furthermore, the consistency of this pattern across ensemble models underscores the robustness of the selected top features and their physical-statistical significance in controlling LST dynamics. Focusing on this compact yet informative set of variables allows for clearer interpretation of environmental drivers of LST variation and facilitates the transferability of models across regions and seasons.

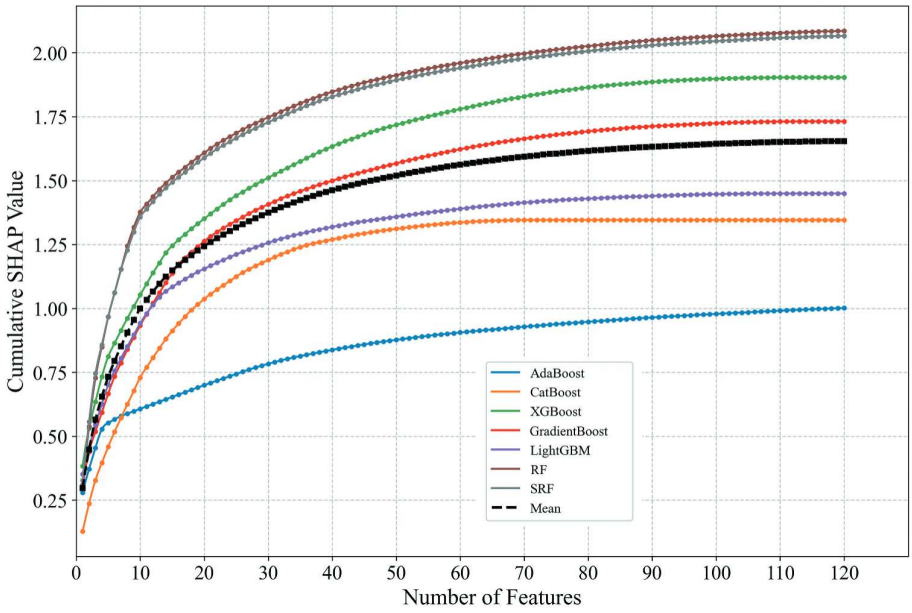


Figure 12. Cumulative SHAP contribution values. The *X*-axis represents the top *x* features ranked by each model, and the *Y*-axis denotes the cumulative contribution values of these top *x* features.

5.2. Feature selection and physical interpretation

The integration of interaction features substantially enhances both model interpretability and LST prediction accuracy by synthesizing multidimensional variables into composite indicators that capture complex, non-additive relationships beyond individual features (Kumar et al. 2023; M'hamdi et al. 2024). SHAP analysis highlights the critical role of these engineered interactions. Among them, the red-blue band combination consistently ranks highest across models such as XGBoost and GradientBoost, exhibiting the broadest SHAP value range (−2.0 to 2.0) (Figure 6). This indicates its dominant influence on LST prediction and its effectiveness in representing surface energy dynamics through visible spectrum reflectance. The red band captures vegetation reflectance, such as chlorophyll absorption (Hunt et al. 2011; Sims and Gamon 2003), especially in dense canopies (Ollinger 2011), while the blue band relates to soil surface moisture (Chen et al. 2018; Narayan 2004; Weidong et al. 2002) and evaporation processes (Metternicht and Zinck 2003), making their combination a comprehensive proxy for vegetation-water-energy interactions (Gueymard et al. 2019; Reddy and Singh 2018). High red and low blue reflect dense, well-watered vegetation, which moderates LST through shading and transpiration, whereas low red and high blue indicate sparse, dry surfaces that accumulate heat (Lee, Park, and Lee 2024; Meyer et al. 2024; Van De Voorde, Vlaeminck, and Canters 2008; Wang, Li, and Chen 2022).

SHAP value asymmetry reveals nonlinear and spatially heterogeneous LST responses. Positive SHAP values correspond to conditions that increase LST (e.g. dry, exposed surfaces), while negative values correspond to cooling effects (e.g. dense, moist vegetation), mapping directly onto local energy balance variations. The vegetation – terrain interaction features, such as EVI_Elevation, EVI_Aspect, and Slope_squared_Aspect_sin, show consistently high importance and concentrated SHAP distributions, indicating stable explanatory power for local microclimates. These interactions integrate biological signals from vegetation with geophysical controls from terrain (Arevalo-Ramirez et al. 2022; Dutta et al. 2024; Piccoli et al. 2023). For example, elevation affects air temperature and humidity, while slope and aspect regulate solar energy input (Kumar, Skidmore, and Knowles 1997); when coupled with EVI, these factors reveal how vegetation moderates surface temperature differently across topographic gradients (Peng et al. 2021; Zhang et al. 2023). This synergy highlights that vegetation vigor and terrain configuration jointly shape energy fluxes and evapotranspiration, thereby exerting strong and spatially stable control over LST.

Across various models, including CatBoost, LightGBM, and SRF, the red–blue combination and terrain–vegetation interactions maintain their top ranking (Figures 5 and 6), demonstrating strong cross-model consistency and adaptability. These findings confirm that the most predictive features for LST estimation are not broad spectral or topographic variables, but rather specific combinations that integrate

vegetation indices and terrain structure. Such stability and explanatory power support their prioritization in model simplification, computational optimization, and large-scale LST inversion. Furthermore, they provide a theoretical foundation for advancing LST downscaling frameworks and bridging the gap between physical surface processes and machine learning-based modeling (Buyantuyev and Wu 2010; Hritz 2014; Li, Zhang, and Huang 2014; Maroni et al. 2021).

5.3. Scale effect of spatial random forest models

This study examines scale effects in spatial random forest LST modeling through multi-resolution analysis (10 m vs 100 m). Performance disparities stem from algorithmic traits, sensor variations, and data heterogeneity. Despite hyperparameter tuning (cross-validation, grid search), 10 m-resolution modeling shows limited performance due to localized noise and environmental intricacy, triggering overfitting risks in high-resolution data processing. Moreover, processing high-resolution data significantly increases computational complexity, leading to longer training times and higher memory consumption, which can limit practical applicability.

High-resolution modeling faces elevated uncertainty (Ren, Mi, and Georgopoulos 2020), as fine-scale predictions are prone to localized environmental noise and micro-pattern-induced overfitting. While SRF exhibits reduced accuracy at 10 m vs 100 m (Figures 8 and 10), its spatial weighting mechanism inherently mitigates errors by prioritizing geographically proximate features, contrasting with conventional noise propagation assumptions. This spatial regularization enhances robustness against resolution-driven complexity. To address computational and stability challenges, potential strategies include feature selection or dimensionality reduction, local/block-wise processing, and parallelized or optimized implementations, which could improve both efficiency and model robustness.

Significant performance differences were observed among different land use types in this study. The stable surface characteristics and pronounced elevation variations in vegetated areas enhanced the model's ability to capture spatial variation patterns and notable elevation changes (Rathjens, Dörnhöfer, and Oppelt 2014). In contrast, barren areas exhibited reduced prediction accuracy due to unstable surface reflectivity and micro-topography complexities. For high-resolution data, incorporating the spatial variability of land use features and local-scale processing techniques could improve the model stability in heterogeneous landscapes. The accuracy of downscaled LST exhibits strong land-cover dependence. Tree-based models (e.g. SRF/RF), which capture nonlinear and higher-order interactions, generally outperform linear baselines, though the degree of improvement varies across ecosystems. Vegetated surfaces (forests, grasslands) demonstrate high accuracy, low error, and stable performance, highlighting the suitability of downscaled LST for drought monitoring, heat anomaly detection, and energy – water balance assessment. However, cropland tends to exhibit greater model sensitivity errors. This variability is not merely due to inherent land cover characteristics; it reflects the limitations in capturing complex temporal dynamics and management-induced variations (such as phenological changes, irrigation effects) through the available predictors. Bare and urban surfaces yield the largest errors and strongest model divergence. This is primarily due to their high spatial heterogeneity, complex surface properties, and strong anthropogenic influences. Urban areas are affected by building geometry, impervious surfaces, and anthropogenic heat emissions, while bare surfaces exhibit rapid temperature fluctuations influenced by soil moisture and albedo variations. Consequently, only models capable of capturing nonlinear interactions among multiple features can achieve significant improvements in predictive accuracy.

6. Conclusions

This study employed machine learning models to downscale Landsat 9 and Sentinel-2 satellite data, generating high-resolution (10 m) surface temperature maps validated against drone measurements. Linear models showed moderate performance at 100 m resolution ($R^2 = 0.58$), while boosting algorithms (XGBoost: $R^2 = 0.79$; CatBoost: $R^2 = 0.75$) and LightGBM ($R^2 = 0.70$) demonstrated enhanced feature utilization. At 100 m resolution, the SRF model achieved optimal accuracy ($R^2 = 0.96$, RMSE = 0.32°C), followed by the standard RF ($R^2 = 0.95$, RMSE = 0.34°C). Although all the model's accuracy declined in 10 m resolution, SRF maintained robust performance in vegetated areas (mean $R^2 = 0.74$, RMSE = 6.28°C). The downscaling process exhibited a clear scale effect. Incorporating spatial autocorrelation (via SRF) effectively mitigated accuracy loss during scale transitions. Significant variations in scale sensitivity among models highlighted the inherent complexity of downscaling thermal data. The proposed framework integrates multi-source remote sensing advantages, while spatially informed modeling significantly improves result reliability.

The feature interaction-optimized machine learning approach establishes a novel paradigm for LST downscaling research, providing a robust analytical framework for understanding the spatial differentiation of thermal environments across diverse regions. However, several limitations remain, including sensitivity to underrepresented land cover types in the

training set, susceptibility to errors induced by fine-scale noise and micro-patterns, and low computational efficiency at high resolutions. Future work could consider balancing training samples across different land cover types to reduce bias toward vegetation-dominated areas, implementing local/block-wise processing or feature dimensionality reduction to improve computational efficiency for high-resolution data, integrating SRF with physically based models to better capture micro-scale thermal processes and enhance robustness in heterogeneous landscapes, and exploring parallelized or optimized implementations for large-area, high-resolution applications.

Author contributions

CRediT: **Wenrui Yuan:** Conceptualization, Methodology, Visualization, Writing – original draft; **Shi Hu:** Funding acquisition, Project administration, Supervision, Writing – review & editing; **Chesheng Zhan:** Project administration, Resources, Writing – review & editing; **Guoqiang Wang:** Funding acquisition, Project administration, Resources; **Youyu Luo:** Data curation, Software.

Disclosure statement

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Data availability statement

The datasets generated and analyzed during the current study can be requested by the corresponding author. All data are available in our public database. The 10 m resolution data generated by downscaling in this study have been deposited at Mendeley (https://doi.org/10.17632/8TKXSBWBKY.1).

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